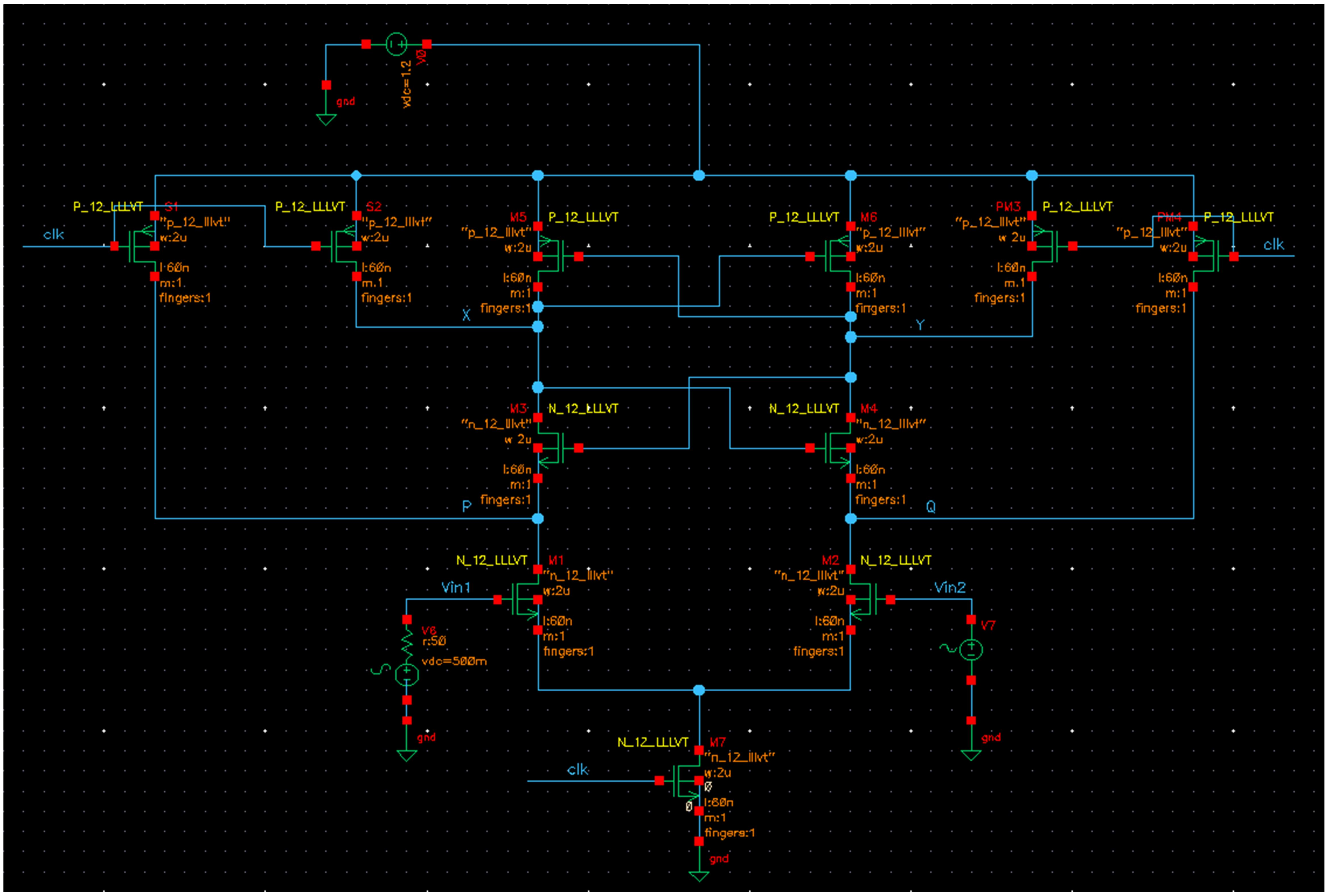
ASSIGNMENT - 2

By – Siddhant Abhyankar

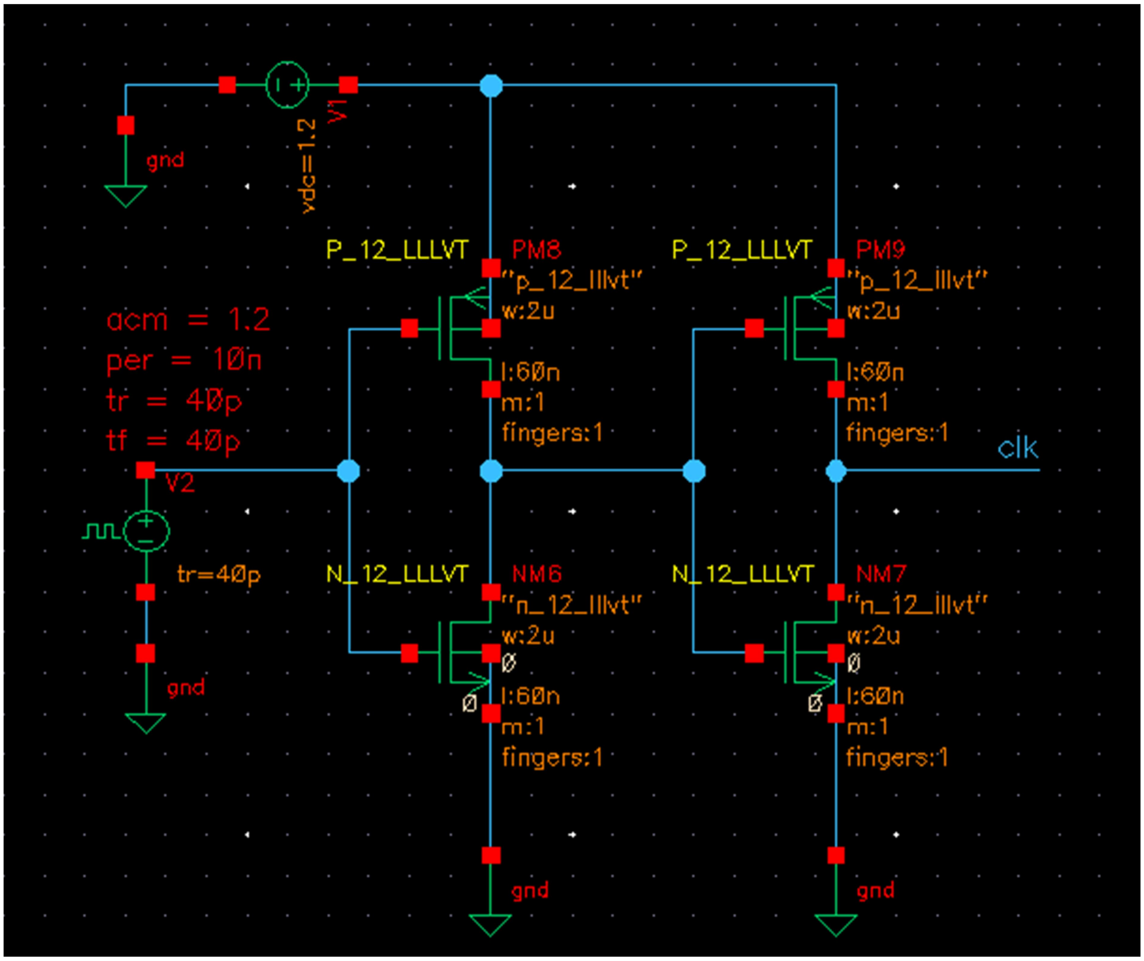
REGISTRATION NUMBER – SC25M085

AIM: Design a strong-arm latch in UMC65nm CMOS process. Add a SR latch at the output so that the output changes for every clock cycle. You can use an ideal clock source with a rise/fall time of 40 ps. Pass this through an inverter chain of 2 inverters before you connect it to your circuit. Clock frequency: 100 MHz.

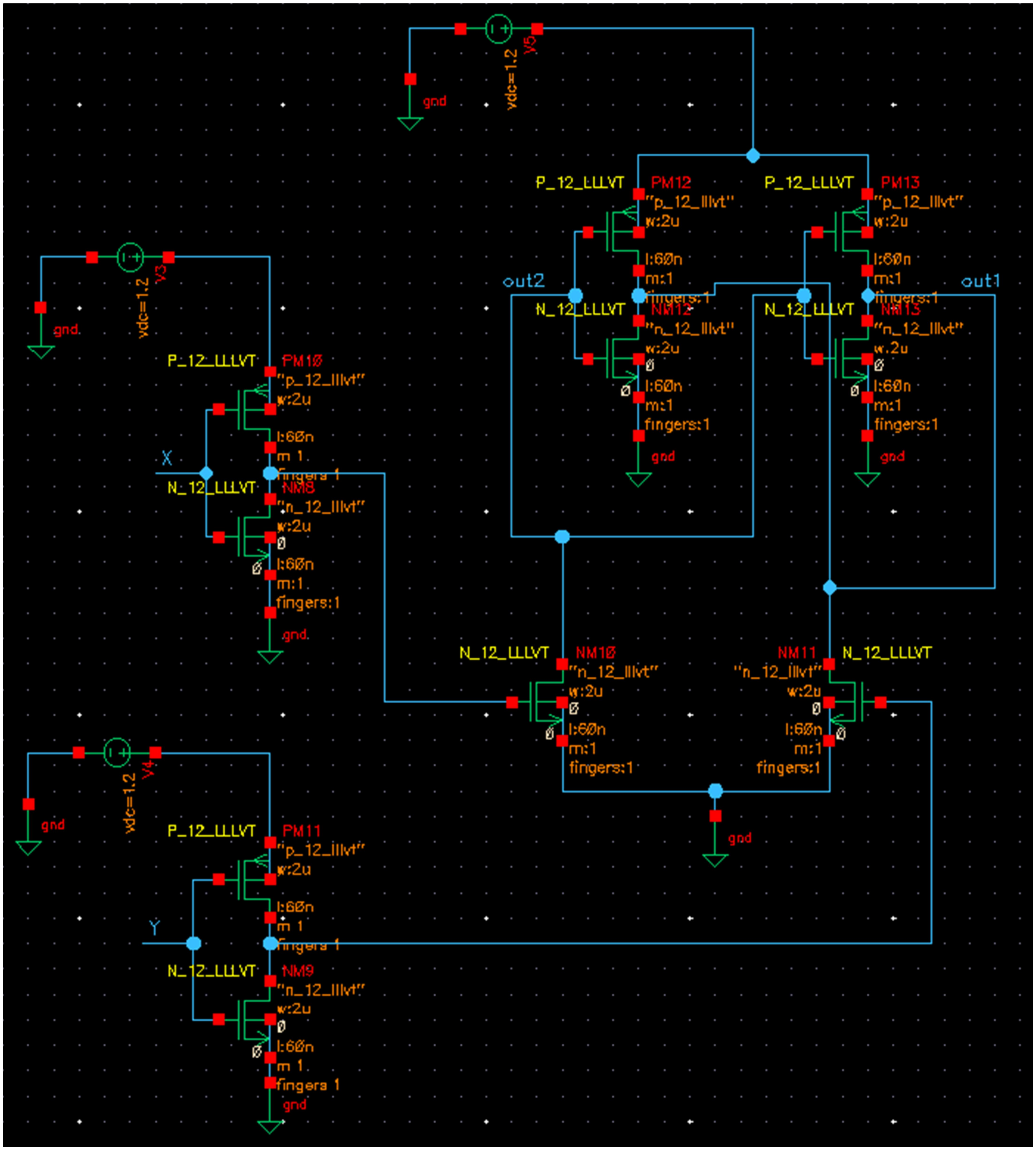
Circuit Dig:



Strong ARM Latch



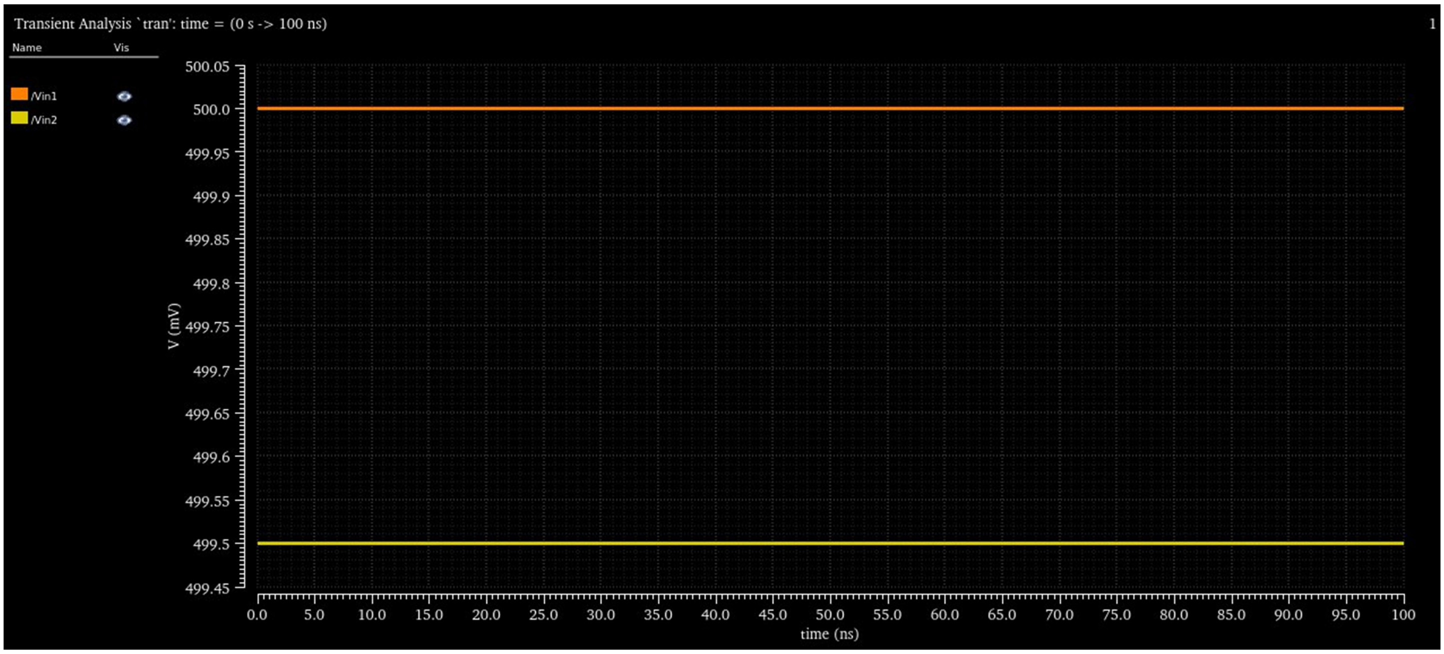
Clock with 2 inverter buffers



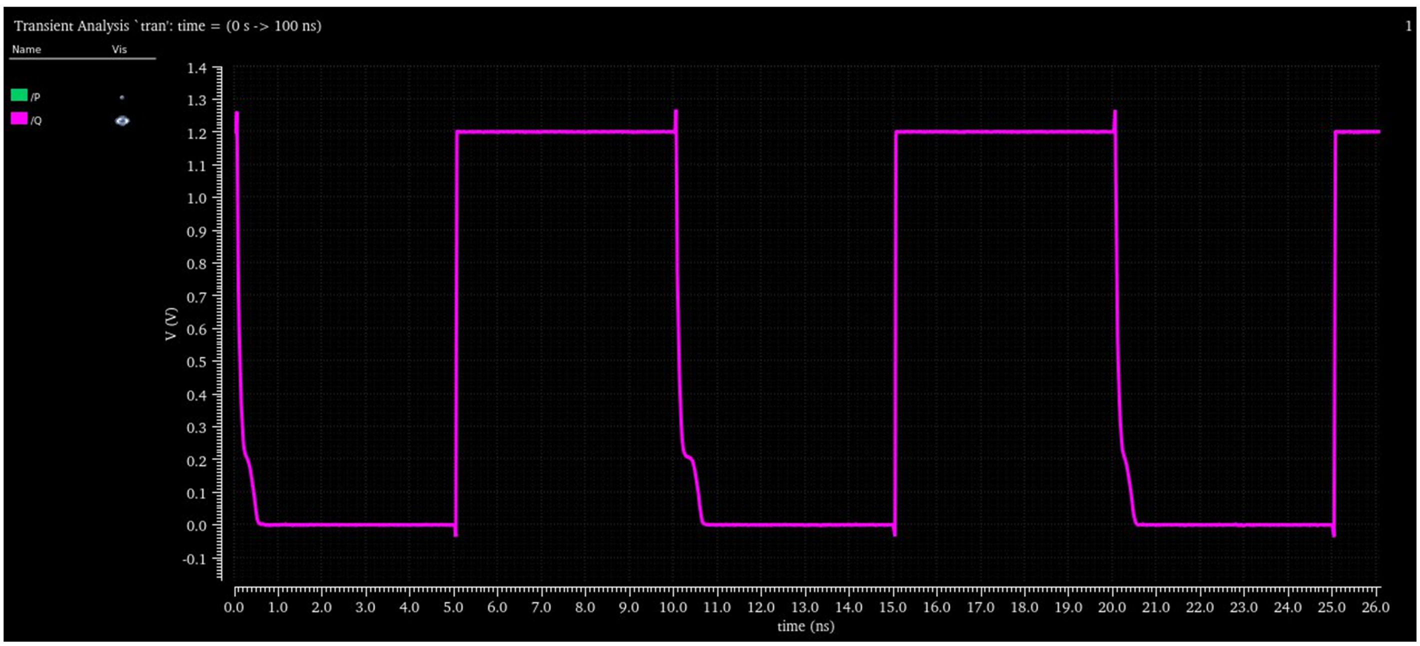
RS Latch for storing the output

Simulations:

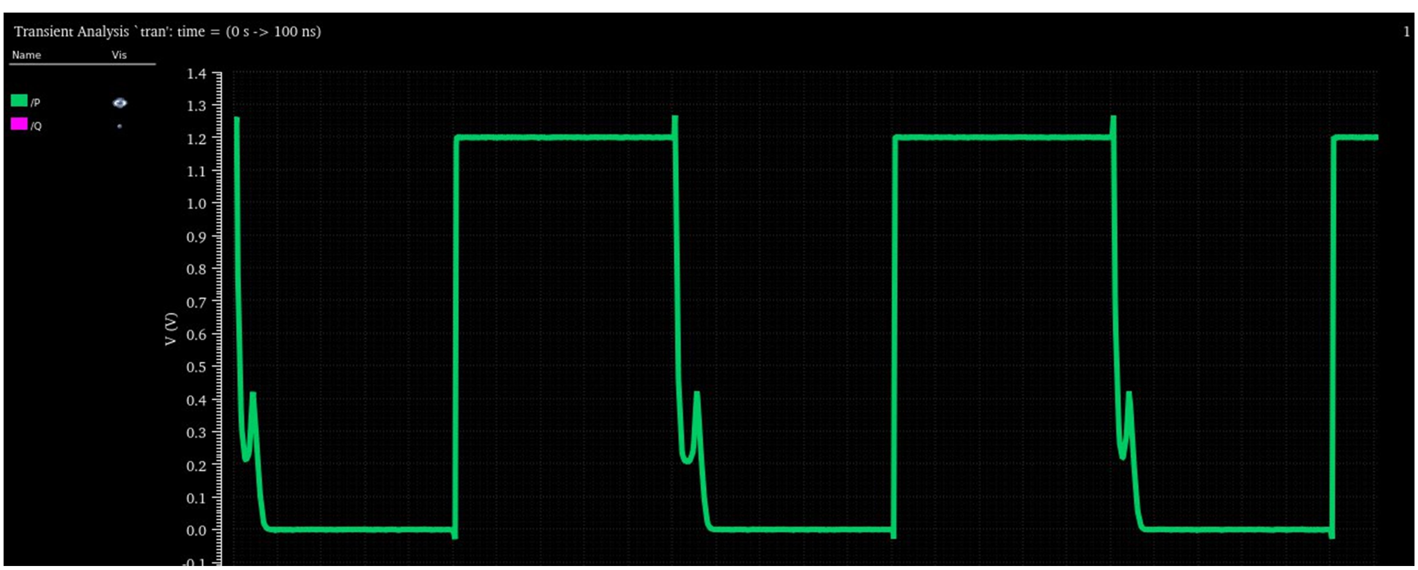
1. *Run transient simulations and show the performance for an input differential voltage of 0.5 mV.*



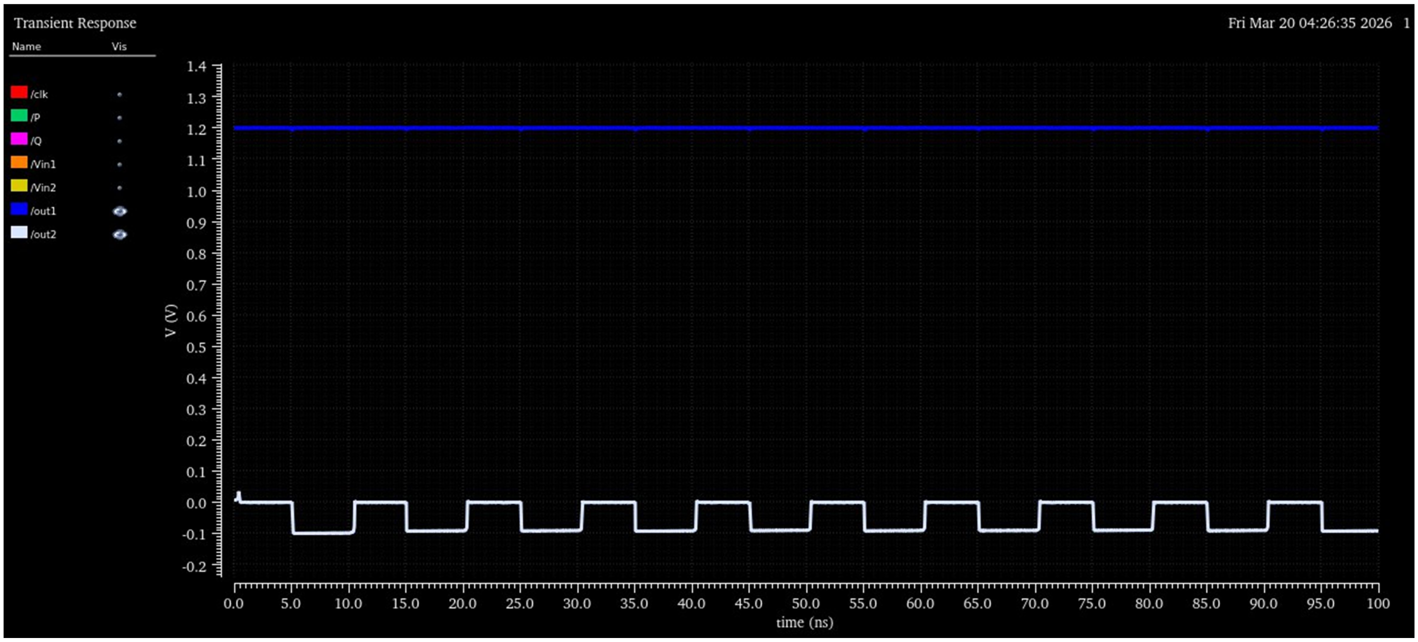
Transient Analysis `tran': time = (0 s -> 100 ns)
/Vin1
/Vin2
V (mV)
time (ns)



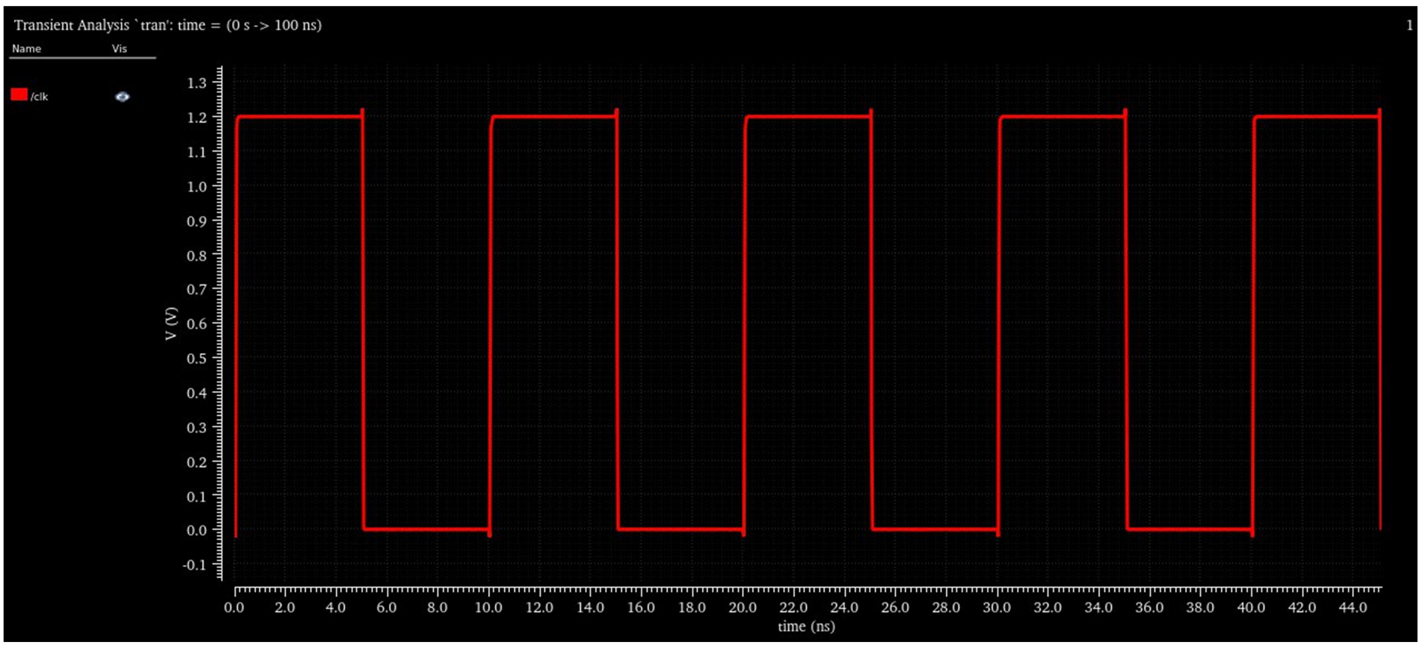
Transient Analysis `tran': time = (0 s -> 100 ns)
/P
/Q
V (V)
time (ns)



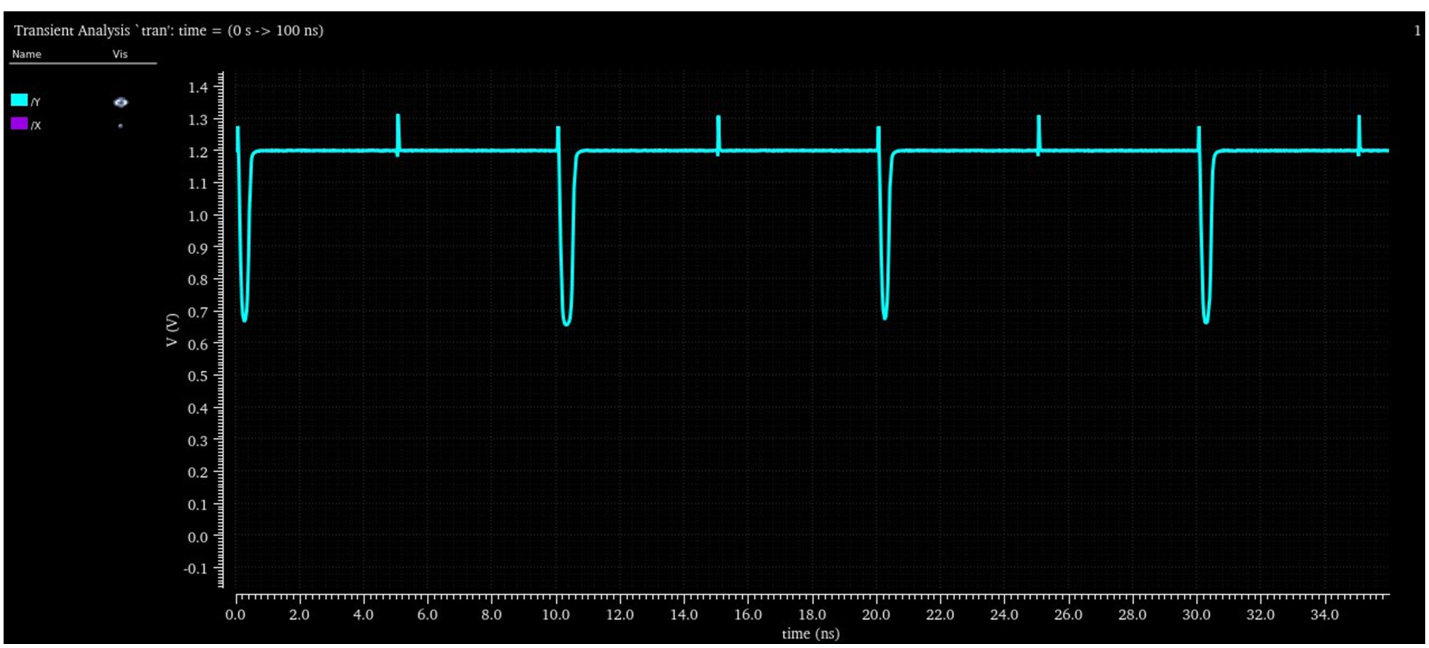
Transient Analysis `tran': time = (0 s -> 100 ns)
/P
/Q
V (V)



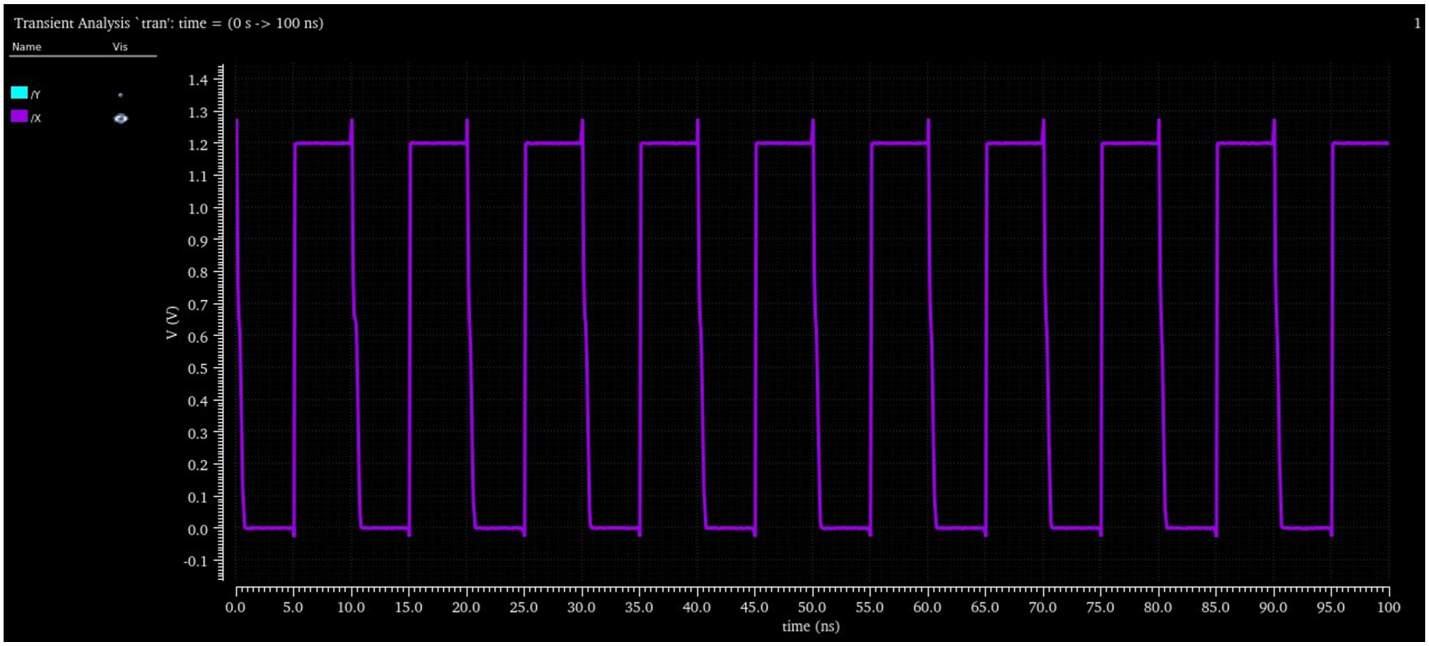
Transient Response
Fri Mar 20 04:26:35 2026 1
Name Vis
/clk
/P
/Q
/Vin1
/Vin2
/out1
/out2
V (V)
time (ns)



Transient Analysis `tran': time = (0 s -> 100 ns)
1
Name Vis
/clk
V (V)
time (ns)

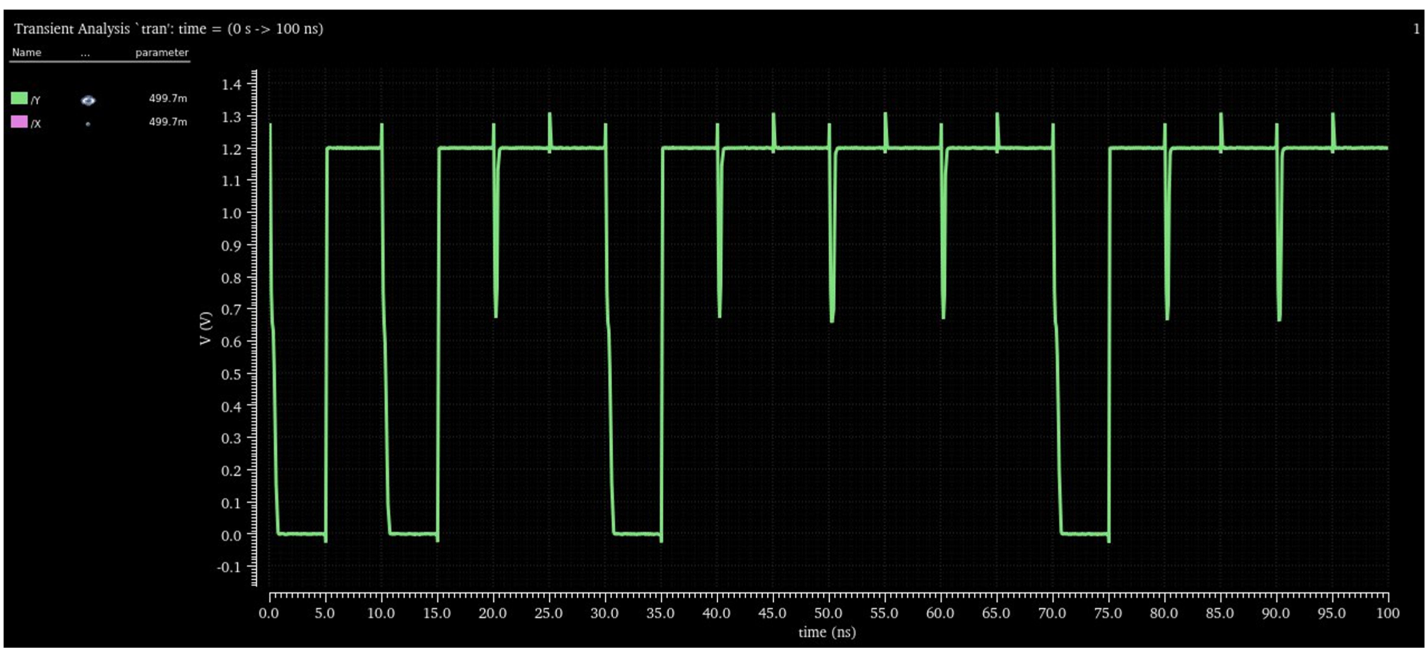


Transient Analysis `tran': time = (0 s -> 100 ns)
1
Name Vis
/Y
/X
V (V)
time (ns)

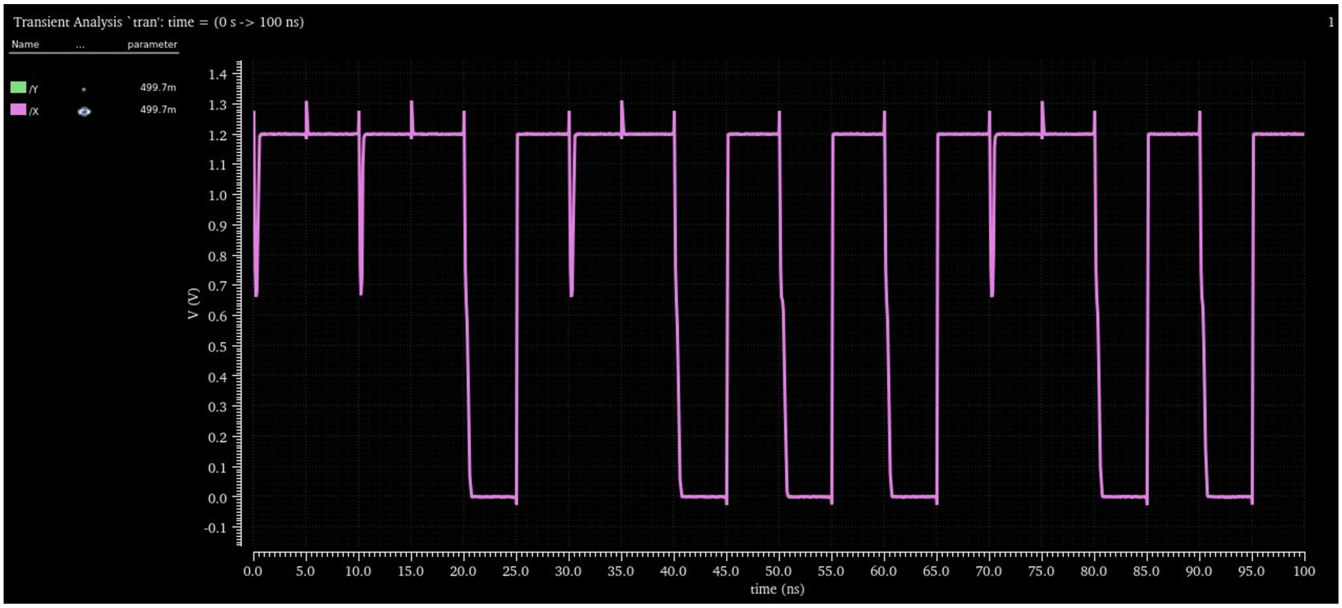


2. *Estimate the lowest input differential voltage that the comparator can resolve, limited by metastability. You can do this by reducing the input differential voltage till the comparator is unable to make a decision.*

Metastability (**where our comparator is unable to make a proper accurate decision about the voltage differences, happens due to noise triggering**) is observed to be on 499.7mV or at a input differential voltage of 0.3 mV

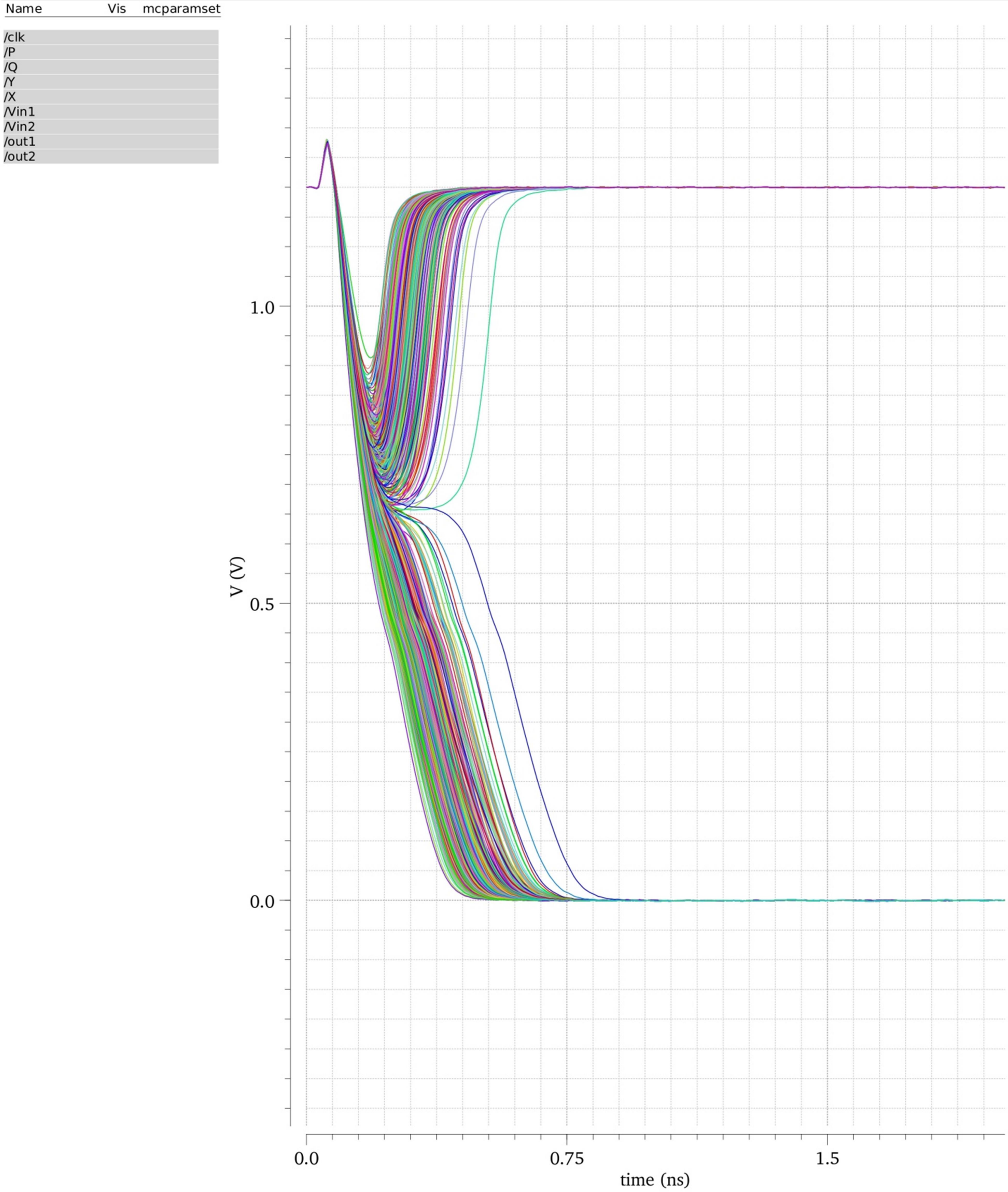


Output at Y was supposed to stay high but since the input voltage differences were very low because of which our strong arm latch went into metastability state. That's why we can here clearly observe that there are multiple occurrences where Y went to zero even it was supposed to not.



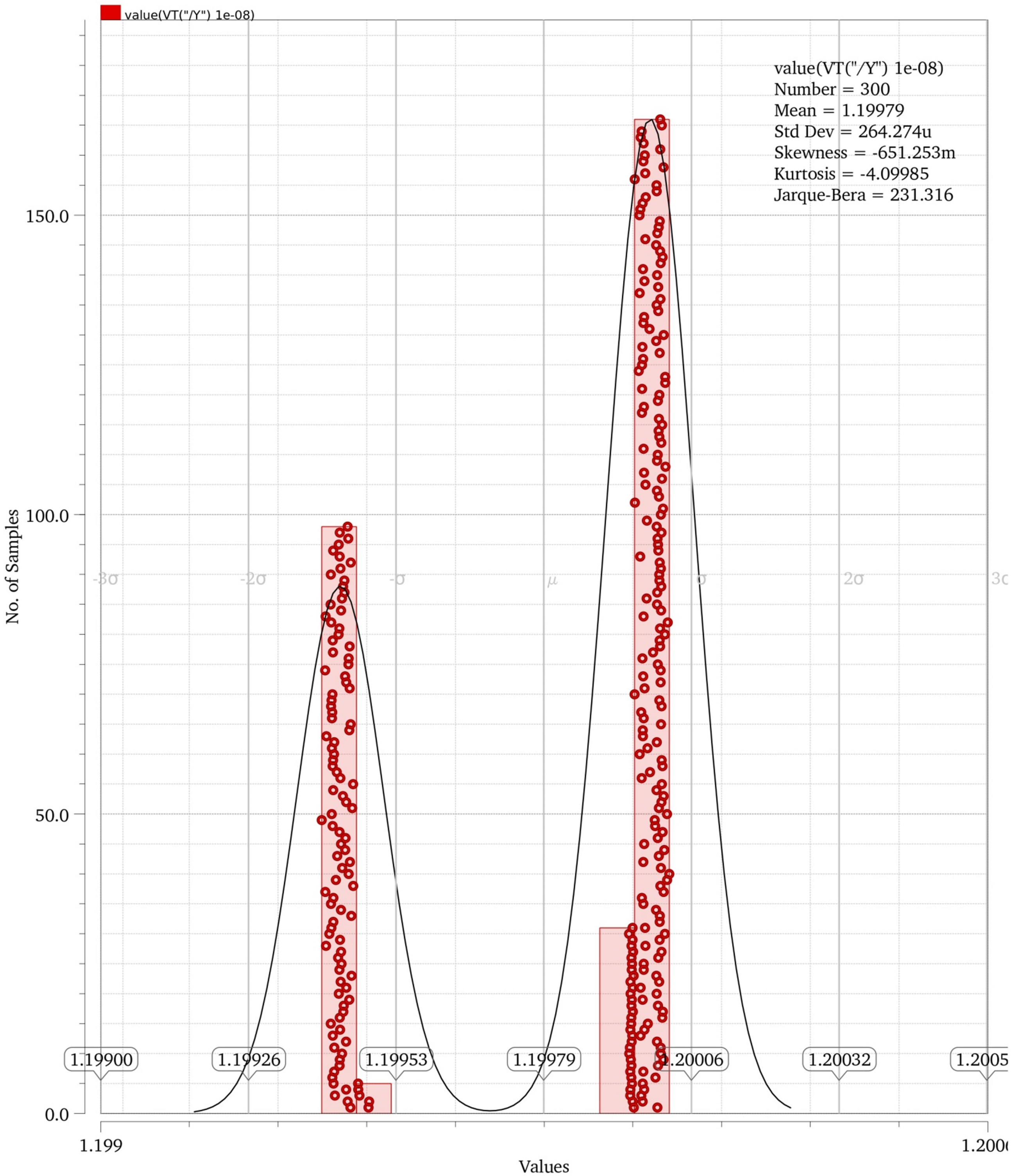
3. *Run Montecarlo simulations (at least 300 runs) and estimate the 3σ of the offset voltage. Plot the histogram.*

Transient Response **Fri Mar 20 05:32:43 2026**



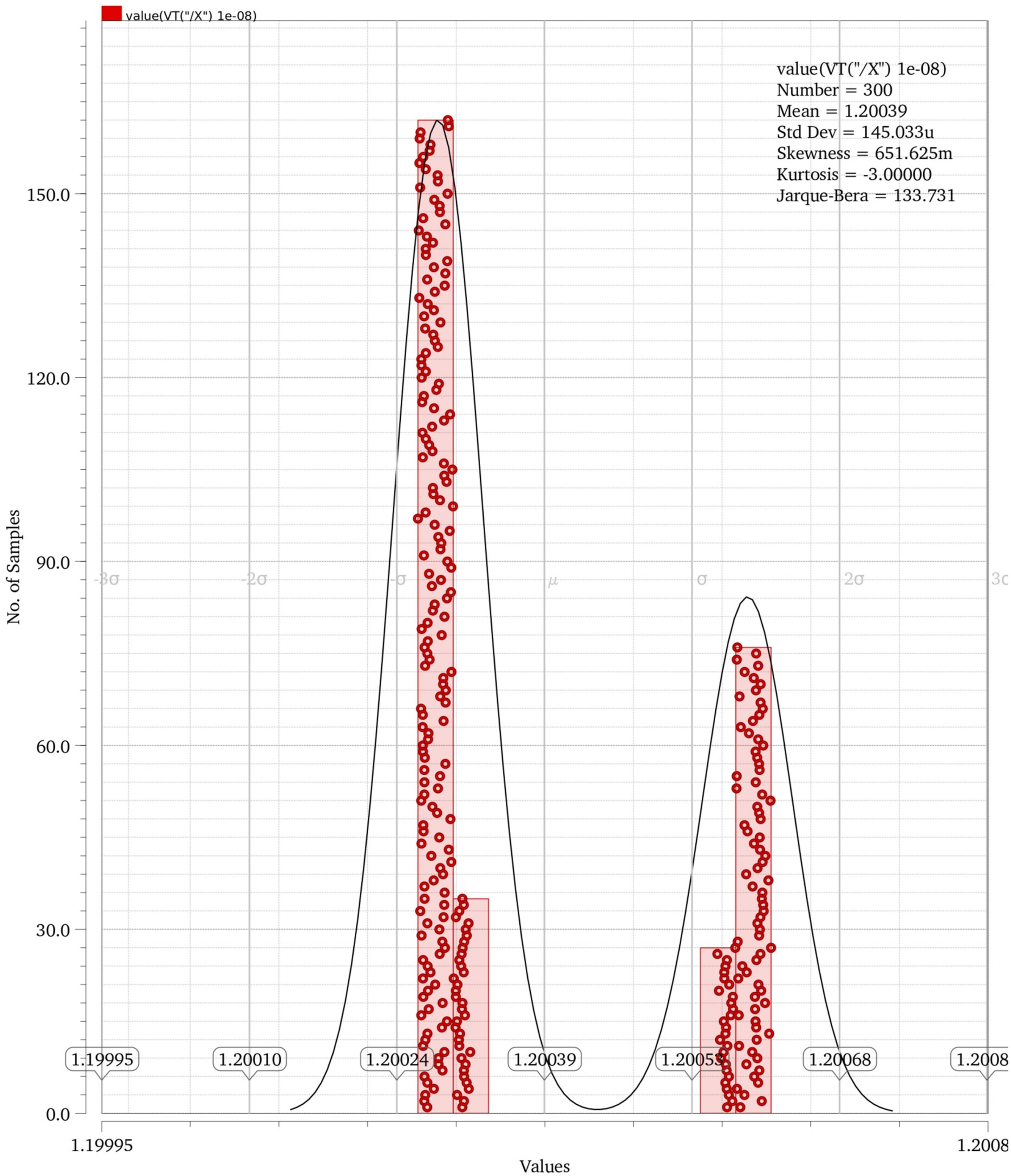
Only X and Y transitions have been showed here

value(VT("/Y") 1e-08)



Here we can clearly see that there are plenty of times where values at Y = 0 i.e. at almost 100 samples even when input at Vin2 was less than Vin1. This behaviour is due random mismatches occur in our circuit after fabrication and montecarlo simulation helps us to reimagine these *mismatches mathematically.*

value(VT("/X") 1e-08)

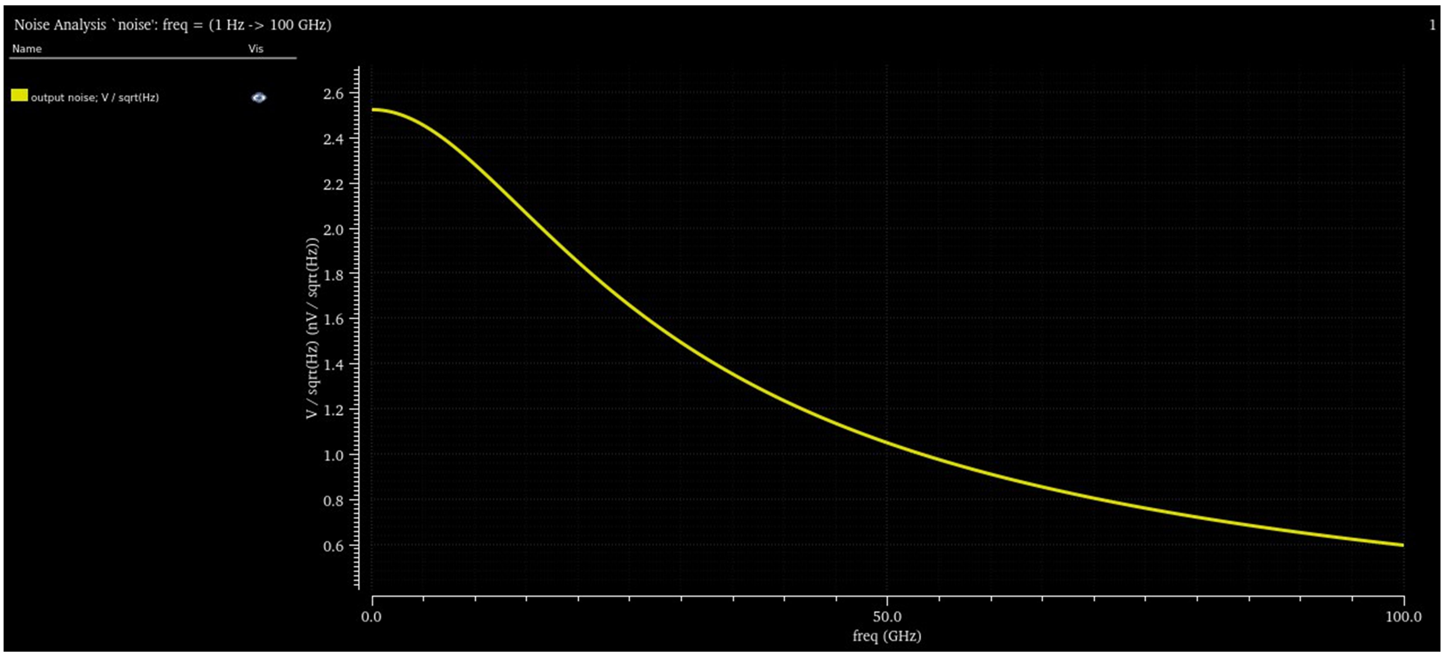


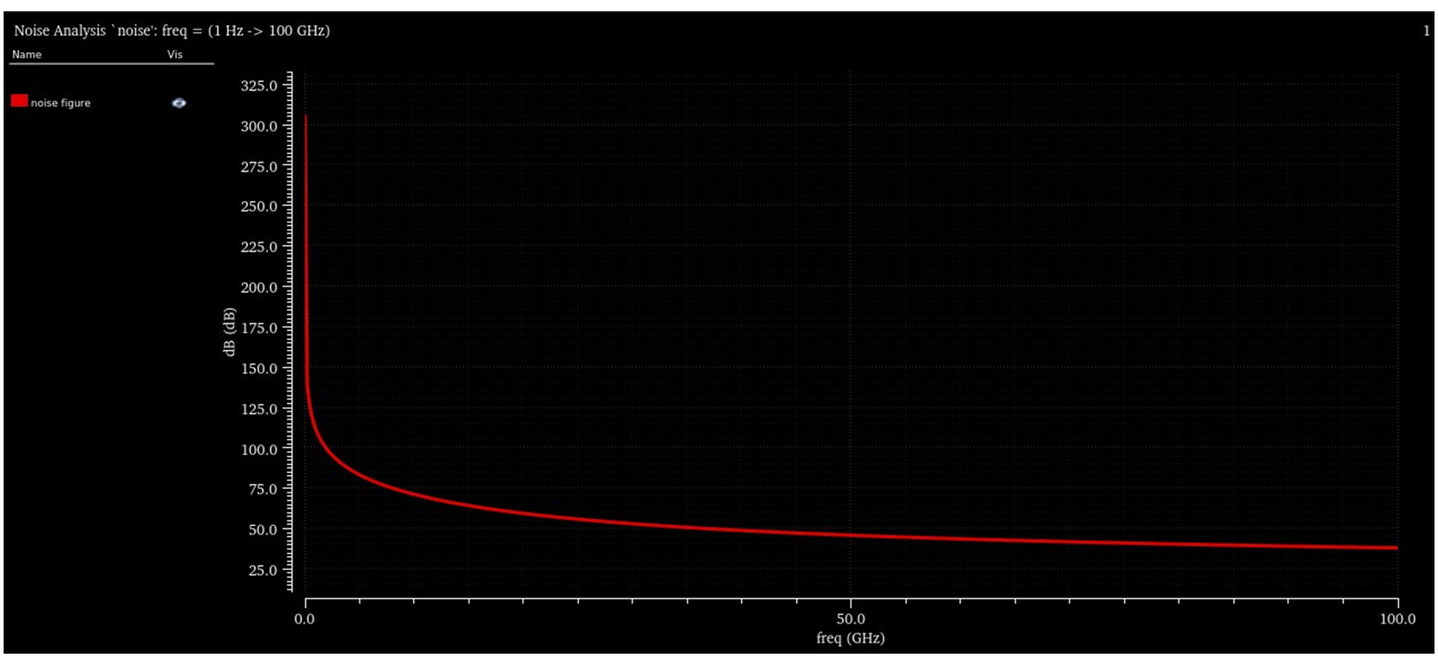
10	value(VT("/X") 1e-08)	1.20017
11	value(VT("/Y") 1e-08)	1.1994

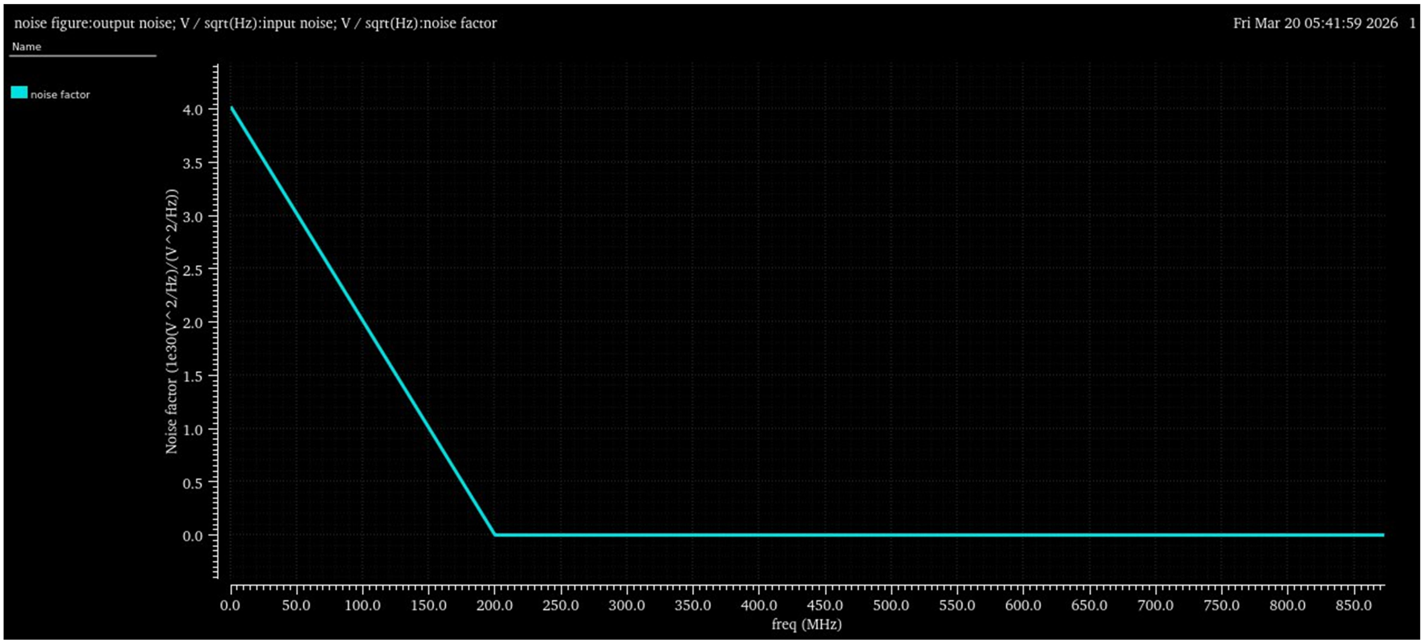
Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Yield Estimate: 100 %(300 passed/300 pts) Confidence Level: <not set> Filter: <not set>									
- analogLib:StrongArmLatch:1									
	- value(VT("/X") 1e-08)(summary)	100% (300/300)	1.2		1.201	1.2	145u		0
	value(VT("/X") 1e-08)	100% (300/300)	1.2	info	1.201	1.2	145u		0
	- value(VT("/Y") 1e-08)(summary)	100% (300/300)	1.199		1.2	1.2	264.3u		0
	value(VT("/Y") 1e-08)	100% (300/300)	1.199	info	1.2	1.2	264.3u		0

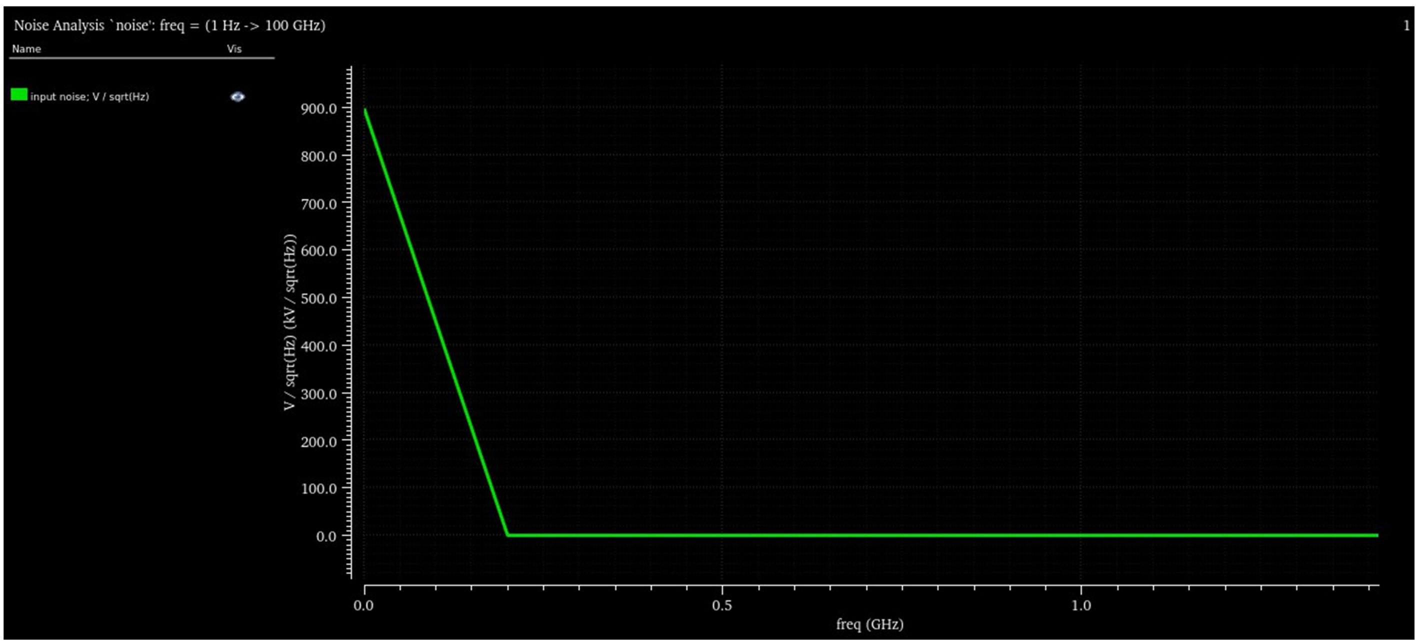
4. *Estimate the noise of the comparator.*

Circuit shows high level of noise figure at lower frequencies than at higher frequencies

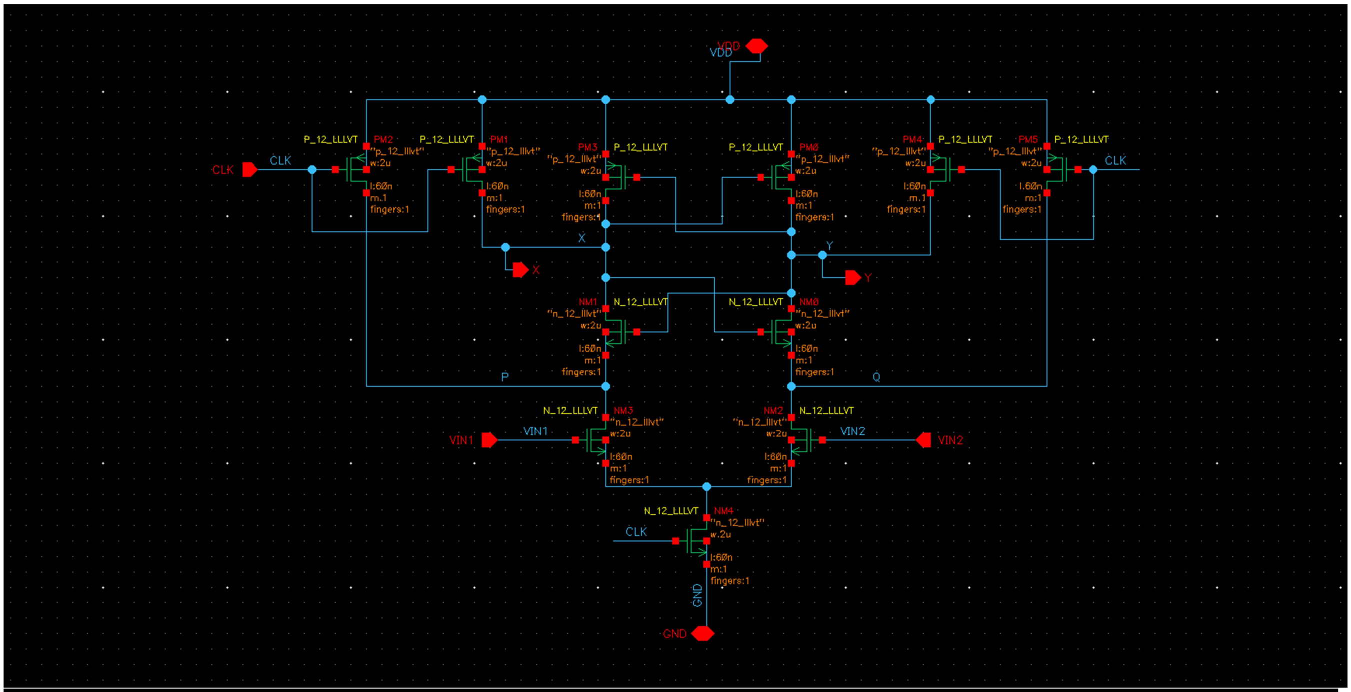


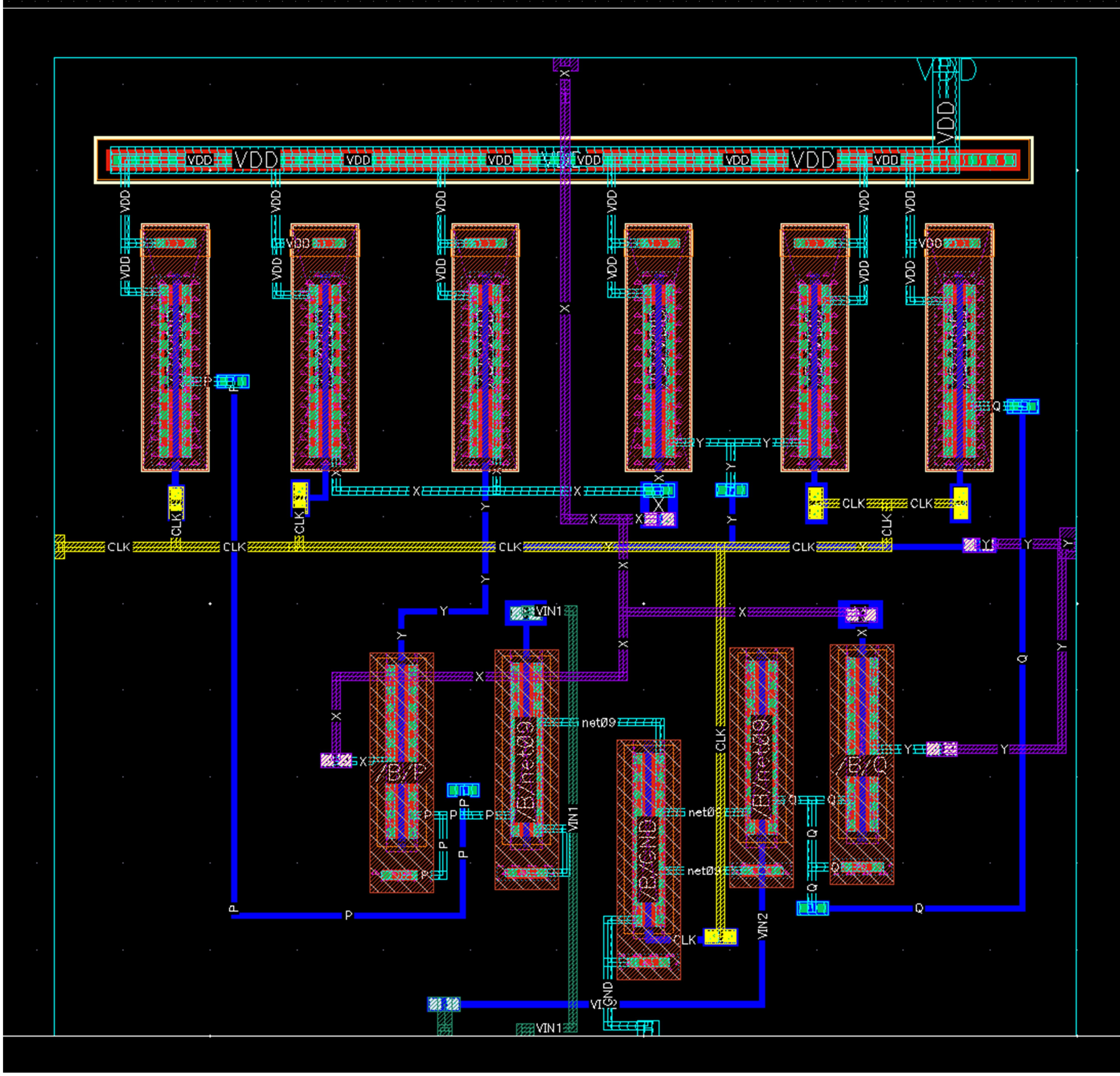



noise figure:output noise; V / sqrt(Hz):input noise; V / sqrt(Hz):noise factor
Fri Mar 20 05:41:59 2026 1
Name
noise factor
Noise factor (1e30(V^2/Hz)/(V^2/Hz))
freq (MHz)


Noise Analysis `noise': freq = (1 Hz -> 100 GHz)
1
Name
Vis
input noise; V / sqrt(Hz)
V / sqrt(Hz) (kV / sqrt(Hz))
freq (GHz)

Layout:


VDD
CLK
P_12_LLLVT
PM2
PM1
PM3
PM0
PM4
PM5
X
Y
NM1
NM0
N_12_LLLVT
P
Q
NM3
NM2
VIN1
VIN2
NM4
GND


VDD
X
Y
CLK
P
Q
VIN1
VIN2
net09
GND